Literature Review and Key Takeaways

Literature Review-

Paper Name & Authors	Brief Synopsis	Workflow / Methodology	Research Gap	Keywords	Importance to Our Project
(Base Paper) **Securing in-vehicle communications through post-quantum cryptography and network segmentation** *(Arcangelo Castiglione & Teresa Elia, 2025)*	Proposes a segmented Controller Area Network (CAN) model integrating Post-Quantum Cryptography (PQC). Uses the CRYSTALS-Kyber Key Encapsulation Mechanism (KEM) to securely share session keys, paired with lightweight symmetric ciphers (RC6, Twofish, AES) to encrypt real-time CAN messages.	1. Segments the vehicle network into functional domains (Powertrain, Chassis, Body & Comfort, Infotainment) managed by a Central Gateway. 2. Utilizes a dedicated "KeyManagement" ECU to generate and distribute the session key at vehicle startup. 3. Fragments the 768-byte Kyber capsule into 12 sequential CAN-FD frames due to the 64-byte payload limit. 4. Evaluates transmission time and bandwidth overhead using a Linux-based C/C++ SocketCAN (vcan) simulation.	• The model does not provide sender authentication, leaving it vulnerable to spoofing and replay attacks. • It cannot mitigate distributed Denial-of-Service (DoS) or malicious gateway (Black/Gray Hole) attacks. • Explicitly recommends the addition of an Intrusion Detection System (IDS) to monitor and block these specific anomalies.	Controller area network protocol Post-quantum cryptography Automotive Network segmentation In-vehicle Security.	Validates the necessity of fragmenting PQC payloads over CAN-FD, directly supporting the Phase 1 indexed block-transfer design. The unresolved vulnerabilities (DoS, spoofing) explicitly justify the implementation of a Gateway-side FL-IDS (Phase 2) and PUF-based ML-DSA remote attestation.
A Survey of Post-Quantum Cryptography Migration in Vehicles *(Nils Lohmiller et al., 2025)*	Provides a structured and comprehensive overview of the current migration state of Post-Quantum Cryptography (PQC) algorithms in the automotive domain. It addresses use cases involving asymmetric	1. Follows a structured five-phase literature research process to analyze and synthesize 60 core papers. 2. Evaluates NIST standardization candidates (CRYSTALS-Kyber, CRYSTALS-Dilithium, Falcon, SPHINCS+) against vehicle constraints. 3. Reviews general	• Identifies 16 specific research gaps (RG1–RG16). • Highlights a lack of crypto-agility in standards and a need for protocol redesigns because current protocols cannot handle large PQC certificate sizes (RG9, RG12). • Notes slower signature verification times with PQC (RG6)	Post-quantum cryptography (PQC) PQC migration vehicles V2V V2X IoV.	Directly justifies our Phase 1 CAN-FD 60/4 indexed block-transfer mechanism, as this paper explicitly proves that standard protocols break under the size of PQC payloads. Furthermore, it validates our Phase 4 Hardware-in-the-Loop (HIL) testing by highlighting the critical lack of PQC performance

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	cryptography across internal vehicle networks, OEM-specific communication, and Vehicle-to-Everything (V2X) communication.	automotive standards (ISO 15118, AUTOSAR, IEEE 1609.2) to assess their crypto-agility and readiness for PQC.	and an absence of performance evaluations on real automotive hardware (RG4).		evaluations on physical automotive silicon.
Deep Learning in the Fast Lane: A Survey on Advanced Intrusion Detection Systems for Intelligent Vehicle Networks *(Mohammed Almehdhar et al., 2024)*	Offers an in-depth examination of advanced machine learning (ML) and deep learning (DL) approaches for developing Intrusion Detection Systems (IDS) to safeguard In-Vehicle Networks (IVNs) against cyber-attacks, specifically focusing on the CAN protocol. It highlights the transition from traditional signature-based to anomaly-based detection methods.	1. Conducts a systematic review of articles published between 2017 and 2024. 2. Proposes a novel taxonomy categorizing IDS into conventional ML, DL, and hybrid models (including Transformers, Federated Learning, and Transfer Learning). 3. Summarizes public benchmark datasets (e.g., HCRL Car-Hacking, ROAD, SynCAN) and simulation tools (e.g., MATLAB, VEINS, CARLA).	• Current DL models are computationally intensive and may not fit on low-end ECUs. • There is a lack of diverse, high-fidelity datasets, as many rely on simulated testbeds with class imbalances. • AI models remain highly vulnerable to adversarial machine learning attacks (like feature poisoning or the fast-gradient-sign method), which can trick the IDS.	In-vehicle network (IVN) intrusion detection system (IDS) machine learning (ML) deep learning (DL) cybersecurity controlled area network (CAN).	Validates our strategy to deploy lightweight FL-IDS models at the Gateway edge rather than pushing computationally heavy DL models directly onto resource-constrained ECUs. It also provides strategic guidance on which datasets (e.g., Car-Hacking) to use for training our local models and executing threat simulations.
Quantum-Resilient Secure Onboard Communication (QRSecOC): Integrating Post-Quantum Cryptography for Robust Automotive Network	Proposes **QRSecOC**, a hybrid cryptographic framework combining classical algorithms (RSA/ECC) with lattice-based PQC to secure	1. Initialization of ECUs & CAN channels. 2. Real-time threat level monitoring and adaptive security level modulation. 3. Hybrid encryption execution (RSA/ECC key exchange + PQC payload protection).	• Tested only via software simulation; lacks validation on physical ECU silicon. • Lacks explicit thresholds/metrics for triggering adaptive security level transitions. • Relies strictly on	Post-Quantum Cryptography QRSecOC Automotive Network Security Lattice-Based Cryptography	Validates the performance advantage of hybrid PQC schemes over standard CAN networks, providing theoretical support for our Phase 1 session wrapping while highlighting why our Phase 4 physical HIL benchmarking is

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Security *(Amjad Nsour, 2025)*	CAN communications between ECUs. Features dynamic, adaptive encryption based on real-time threat levels.	4. Multi-objective optimization balancing latency, computational overhead, and energy consumption.	lattice cryptography without exploring alternative PQC paradigms.	ECU Security, CAN.	essential.
A Survey of Post-Quantum Cryptography Migration in Vehicles *(Nils Lohmiller et al., 2025)*	Provides a comprehensive survey mapping the migration state of PQC across four automotive domains: General Automotive Standards, Internal Vehicle Networks, OEM-Specific Communication, and V2X.	1. Systematic literature review across 60+ core studies. 2. Evaluation of automotive security standards (ISO 15118, AUTOSAR SecOC, ETSI TS 103 097, IEEE 1609.2). 3. Classification of IVN bus protocols (CAN, CAN-FD, FlexRay, Ethernet) and ECU hardware platforms. 4. Mapping 16 research gaps (RG1–RG16) to proposed migration strategies.	• Highlights 16 distinct gaps, including lack of crypto-agility in standards, protocol breakdown due to large explicit PQC certificate payloads, absence of side-channel attack (SCA) analyses on automotive PQC, and lack of gateway retrofits for legacy vehicles.	Post-quantum cryptography (PQC) PQC migration vehicles V2V V2X IoV.	Serves as a foundational reference confirming that large PQC payload sizes cause packet fragmentation, directly justifying our Phase 1 60/4 indexed block-transfer mechanism for large attestation objects.
Quantum Federated Learning for Autonomous Vehicle Cybersecurity: An Analytical Review of Architectures and Threat Landscapes *(Sowmya T & Raghavendra M Devadas, 2026)*	Introduces a 4-layer Quantum Federated Learning (QFL) reference architecture for AV Intrusion Detection Systems (IDS) combining FL, CRYSTALS-Dilithium gradient authentication, QKD, and Quantum Krum Byzantine filtering.	1. Multi-dimensional risk matrix analyzing AV threat vectors across 46 studies. 2. Comparative evaluation of 10 FL-IDS studies on accuracy vs. convergence rounds. 3. Technology Readiness Level (TRL) ranking of quantum transition paths. 4. Formulation of a 5-stage QFL pipeline and a 2025–2030 development roadmap.	• "Simulation-to-reality deficit" (no QFL validated on physical automotive hardware or NISQ devices). • Lack of quantum Byzantine fault tolerance scalable to 1000+ fleets. • Absence of a standardized benchmark suite (**AV-SecBench**). • Regulatory lag in standards like ISO/SAE 21434.	Autonomous vehicles Cybersecurity Federated Learning Quantum Key Distribution Post-Quantum Cryptography Quantum Federated Learning Intrusion Detection	Supports our Phase 2 Gateway-side FL-IDS strategy, confirming that local edge inference using LSTM Autoencoders combined with Flower (flwr) aggregation protects privacy while mitigating central server vulnerabilities.

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				CAN Bus Security.	
PQ-CAN: A Framework for Simulating Post-Quantum Cryptography in Embedded Systems *(Mauro Conti et al., 2025)*	Introduces **PQ-CAN**, an open-source, Docker-based containerized simulation framework designed to evaluate PQC algorithms (KEMs and DSAs) under constrained CAN bus 2.0 conditions.	1. Docker containerization of ECUs communicating over virtual CAN (vxcan) interfaces with CPU throttling (120/200/300 MHz) and bus bitrates (125k/500k/1M). 2. Injection of background traffic via canbusgen loading the bus to 88% capacity. 3. Benchmarking 7 NIST PQC algorithms (Kyber, BIKE, HQC, McEliece, Dilithium, SPHINCS+, Falcon). 4. Performance evaluation across latency, success rates under 2s timeouts, and communication overhead.	• Relies on Docker CPU throttling on a Linux host rather than physical silicon ECUs. • Evaluates standard CAN 2.0 (does not test CAN-FD 64-byte frame payloads or CAN-XL). • Benchmarks software C implementations without hardware cryptographic acceleration.	Post-Quantum Cryptography Embedded Systems Automotive CAN bus.	Provides the exact software tooling methodology (Docker + SocketCAN vcan + can-utils) adopted for our Phase 1 & 2 co-simulation environment, while proving why upgrading to CAN-FD is required for PQC payloads.
Post-Quantum Cryptography for ECU Security Use Cases *(Fabio Campos et al., 2019)*	Conducts a TARA across core ECU use cases and benchmarks post-quantum digital signature schemes (Dilithium and XMSS) against EdDSA for secure Over-The-Air (OTA) firmware flashing on embedded microcontrollers.	1. STRIDE-based TARA mapping risk levels across 4 basic ECU use cases (Software Update, Diagnosis, Secure Boot, Onboard Comm). 2. Pre-selection of candidate signature schemes meeting memory and speed limits. 3. Hardware testbed using a Raspberry Pi 3B (Backend) flashing firmware via UART to an ARM Cortex-M4 @ 168 MHz (Target ECU).	• Evaluates code execution on a generic ARM Cortex-M4 CPU without hardware security modules (HSMs/EVITA). • Stateful hash-based signatures (XMSS) suffer from key-state management issues during frequent updates. • Does not evaluate real-time CAN bus frame constraints.	Automotive Domain Electronic Control Unit Post-Quantum Cryptography Secure Software Update.	Directly validates our choice of ML-DSA (Dilithium) for Phase 1 remote attestation, proving that Dilithium verification achieves a superior cycle-per-byte efficiency (2,305 cycles/byte) compared to classical EdDSA.

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		4. Benchmarking verification cycles, execution time (ms), and cycles/byte.			
Intrusion Detection System for Controller Area Network *(Vinayak Tanksale, 2024)*	Develops an intra-vehicle CAN IDS featuring an LSTM-based anomaly detection engine combined with a multi-parameter Decision Engine to resolve contradictory anomaly outputs using contextual data.	1. Matrix formulation of CAN variable relationships based on mechanical/electrical dependencies. 2. Function design algorithm enforcing graph properties (Hamiltonian cycles, degree ratios, artificial edge bounds). 3. LSTM networks predicting function values ({y}_t) vs. computed values (y_t) to flag anomalies. 4. Decision Engine calculating Evidence Scores (ES_{ik}), Jaccard distance (d), and running a pairwise RESOLVE algorithm. 5. Validation on real vehicle datasets (Toyota, Kia, Opel, Chrysler) under DoS, Spoofing, and Fuzzy attacks.	• Evaluated on limited vehicle datasets without representation from all major OEMs. • Static function rules could be circumvented if an attacker learns the variable relationship matrix (lacks dynamic periodic rule rotation). • Does not integrate cryptographic framing or payload encryption.	Controller area network Deep learning Intrusion detection system Long short-term memory Machine learning Recurrent neural networks.	Directly informs our Phase 2 Gateway-side FL-IDS feature engineering and LSTM Autoencoder design, showing how time-series feature extraction prevents benign vehicle maneuvers from triggering false positives.
Intrusion Detection Systems for In-Vehicle Networks: Protocols, Applications, and Challenges *(Junaid Ahmad Khan et al., 2025)*	A survey reviewing IVN protocols (CAN, CAN-FD, CAN-XL, LIN, FlexRay), cross-industry CAN applications (Automotive, Marine, BMS, Aerospace, Agriculture), and IDS classification	1. Protocol specification analysis and security weakness mapping. 2. Cross-domain examination of CAN bus utilization across vertical industries. 3. Systematic categorization of IDS methodologies (Physical fingerprints, Parametric timing, Entropy, Deep	• Scarcity of standardized, open-community real-world datasets. • Heavy LLM/Transformer-based IDSs exceed the memory/compute envelopes of embedded ECUs for real-time execution. • Lack of hybrid IDS implementations	Automotive security in-vehicle network (IVN) intrusion detection systems (IDS) quantum safe security.	Justifies our Phase 2 choice of exporting lightweight LSTM Autoencoders to ONNX for edge inference, ensuring our Gateway IDS stays well within the strict 10ms real-time ECU execution budget.

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	paradigms.	Learning, LLMs). 4. Identification of technical deployment constraints (real-time scheduling, hardware limits, V2X dynamics).	combining physical-layer security with lightweight edge machine learning.		
Lightweight Cryptography and Physically Unclonable Functions for Secure In-Vehicle Communications *(Arcangelo Castiglione et al., 2025)*	Proposes an in-vehicle security protocol combining SRAM-PUFs for ECU mutual authentication with Lightweight AEAD (ASCON vs. AES-GCM) and PQC for session key distribution over CAN.	1. Enrollment Phase: FOB device collects challenge-response pairs from ECU SRAM-PUFs to build a response hash/key database. 2. Authentication & Key Distribution: Gateway and FOB perform PQC-authenticated handshake; Gateway broadcasts challenge; ECUs respond; FOB verifies response hashes and distributes encrypted symmetric session keys. 3. Secure Message Transmission: Operational CAN messages encrypted/decrypted via Lightweight AEAD (ASCON / AES-GCM). 4. Formal security verification using ProVerif.	• Formal verification in ProVerif revealed a potential Man-in-the-Middle vulnerability where intercepted PUF response hashes could expose session keys if key rotation is disabled. • Lacks physical hardware implementation and latency/power benchmarking on real microcontrollers.	Physically Unclonable Functions Post-Quantum Cryptography CAN protocol In-Vehicle Security.	Serves as the direct inspiration for our Phase 1 core architecture (SRAM-PUF emulation + ML-DSA attestation + HKDF key derivation + ASCON-AEAD session wrapping), mathematically proving our $2n$ message complexity and frame allocation.
Securing the Road Ahead: A Survey on Internet of Vehicles Security Powered by a Conceptual Blockchain-Based Intrusion	Investigates IoV security challenges and proposes a conceptual multi-layered, blockchain-enabled Intrusion Detection System	1. Systematic literature review covering IoV threats, communication layers (V2V, V2I, V2P, V2G), and security standards. 2. Hybrid Deployment Model: Fog layer handles data capture/normalization;	• High transaction latency and sharding overhead of public blockchains create severe bottlenecks for real-time safety-critical V2X messaging. • Lacks empirical validation on physical vehicular	artificial intelligence blockchain internet of vehicles intrusion detection systems	Provides architectural context for our Phase 2 multi-vehicle V2X fleet simulation, highlighting why privacy-preserving local edge inference at the Gateway is superior to heavy

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Detection System for Smart Cities *(Fatima Al-Quayed et al., 2025)*	integrating AI, fog/edge computing, and smart contracts.	Edge layer executes feature extraction (PCA) and K-Means clustering. 3. Blockchain & Smart Contracts: Immutably log security telemetry and trigger automated responses (isolation, network reconfiguration). 4. Mathematical threat modeling and time complexity proofs.	edge devices or live CAN-FD testbeds. • High computational cost of executing continuous smart contracts at the vehicle level.	smart cities smart contracts.	blockchain ledgers for real-time CAN bus protection.

Key Takeaways for Architecture

1. **Protocol Selection (Phase 1):** The literature explicitly confirms that classical CAN 2.0 (8-byte payload) is completely saturated by PQC key exchange and signature data. Moving to CAN-FD (64-byte payload) combined with our 60/4 indexed block-transfer protocol is the optimal path for ML-DSA attestation.
2. **Cryptographic Suite (Phase 1):** Integrating SRAM-PUFs with ML-DSA for remote attestation and ASCON-AEAD for runtime frame wrapping achieves post-quantum resilience and minimal message overhead (2n complexity).
3. **Gateway FL-IDS Design (Phase 2):** To meet real-time automotive execution constraints (<10ms latency), exporting an LSTM Autoencoder to ONNX for local edge inference is far more practical than deploying resource-intensive LLMs or public blockchain smart contracts.
4. **Hardware Validation (Phase 4):** Nearly every paper reviewed cites a lack of physical hardware validation as its primary limitation. Our planned Raspberry Pi + MCP2518FD SPI-to-CAN + INA219 current sensor testbed directly solves this major research gap.